

Supply Chain Analysis and Intelligence for Pharmacy V13 Platform

B. M. Osman*

September 8, 2026

Abstract

Pharmacy V13 is conceived as an integrated management intelligence system that transforms routine pharmacy operations into structured, evidence-based decision support. Rather than functioning solely as a system for recording medicines, purchases, sales, inventory, suppliers, and financial transactions, V13 establishes a progressive analytical architecture linking operational evidence to information, analytical understanding, management intelligence, and accountable decision-making.

The system incorporates a dedicated Supply Chain Analysis and Intelligence Layer, supported by the REEM intelligence engine, to integrate observed purchasing, sales, inventory, supplier, expiry, and demand evidence. Detailed computational outputs from analytical models are transformed into management summaries, visual analytics, risk and exception indicators, model-readiness assessments, and evidence-traceable management signals. The analytical portfolio includes demand forecasting, stock coverage, ABC analysis, expiry risk, reorder point, safety stock, economic order quantity (EOQ), and Monte Carlo uncertainty analysis.

A central principle of the architecture is the distinction between observed evidence, derived indicators, model outputs, and decision intelligence. Model execution alone is therefore not considered sufficient for operational use. Models are assessed according to their analytical purpose and evidence requirements. Continuous forecasting models are evaluated through forecast error

*Al Nahada University, Cairo
babikerosman@yahoo.com

and residual analysis, optimisation models through parameter and sensitivity analysis, while appropriate binary classification models may be evaluated using confusion matrices, precision, recall, sensitivity, specificity, F1 score, calibration, and receiver operating characteristic (ROC) analysis with area under the curve (AUROC), where adequate labelled historical outcomes exist.

The architecture adopts a progressive model-readiness framework ranging from *Available* and *Partial* to *Ready for Extension* and *Operationally Validated*. This prevents mathematically completed calculations from being interpreted automatically as validated procurement recommendations. Instead, Pharmacy V13 positions analytical models as transparent instruments for management understanding, risk assessment, and decision support.

The overall objective is to establish a scientifically defensible and evidence-traceable pathway from observed pharmacy transactions to reliable historical knowledge, validated analytical models, risk-aware prediction, responsible optimisation, and accountable management action. In this architecture, artificial intelligence and statistical modelling inform management; they do not replace human judgement or organisational responsibility.

Contents

1	Supply Chain Analysis and Intelligence	3
1.1	Overview	3
1.2	Supply Chain Intelligence Architecture	3
1.3	REEM as the Supply Chain Intelligence Layer	3
1.4	Evidence Status	4
1.5	Management Summary Rather Than Raw Tables	5
1.6	Supply Chain Visual Analytics	5
1.7	Analytical Model Portfolio	6
1.8	Model Assessment and Validation	7
1.9	ROC Curve and Classification Assessment	8
1.10	Model Readiness	9
1.11	Management Interpretation	10
1.12	Supply Chain Management Cockpit	11
1.13	Future Development	11

1 Supply Chain Analysis and Intelligence

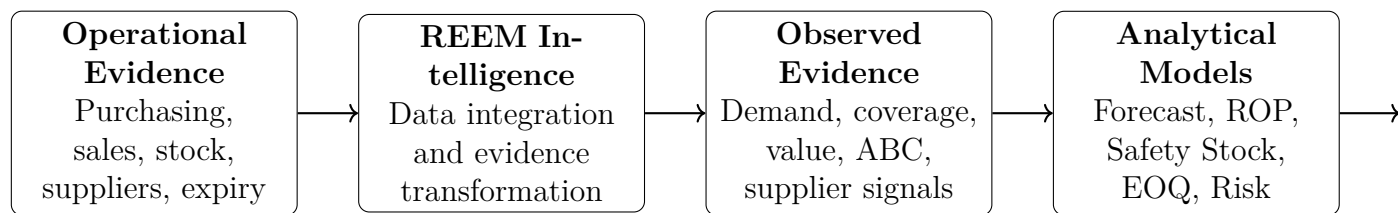
1.1 Overview

Pharmacy V13 extends its management intelligence architecture through a dedicated **Supply Chain Analysis Layer**. This layer connects operational purchasing, sales, inventory, supplier, expiry, and demand evidence with analytical models and management interpretation.

The purpose is not to reproduce long model tables. Instead, the system transforms detailed computational outputs into concise management signals, visual summaries, analytical diagnostics, and model-assessment information.

1.2 Supply Chain Intelligence Architecture

The conceptual flow is:



1.3 REEM as the Supply Chain Intelligence Layer

REEM provides the analytical bridge between the operational V13 database and management-facing supply chain intelligence.

Its conceptual responsibilities include:

- integrating purchasing, sales, inventory, supplier, and expiry evidence;
- determining the available observation period;

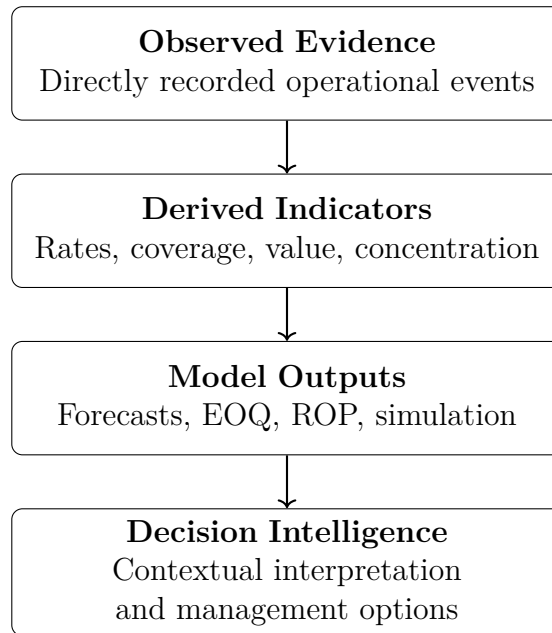
- measuring observed demand from actual transactions;
- calculating inventory coverage and exposure;
- generating preliminary observed-sales ABC analysis;
- summarising supplier activity and procurement cost;
- identifying data-quality exceptions;
- assessing readiness for more advanced analytical models.

The REEM layer therefore establishes an important distinction between **what the pharmacy has actually observed** and **what an analytical model may estimate about the future**.

1.4 Evidence Status

Supply chain intelligence should explicitly communicate the maturity of its evidence.

A conceptual evidence hierarchy is:



The system should never present a model output with greater certainty than is supported by the underlying evidence.

For example, an EOQ calculation may be mathematically completed while the model remains only partially validated because ordering and holding-cost parameters or sufficiently mature demand history are unavailable.

1.5 Management Summary Rather Than Raw Tables

Individual Python models may generate large tables containing results for many medicines. These tables remain useful as computational and audit artefacts, but they should not be the primary management interface.

The management layer should instead provide:

- key aggregate indicators;
- ranked exceptions;
- top-risk medicines;
- concentration and distribution summaries;
- concise model-status information;
- visual trends and comparisons;
- model assumptions and evidence status;
- access to detailed analytical results when required.

The conceptual transformation is:

Large Model Output → Summary → Visual Pattern → Interpretation → Management Attention.

1.6 Supply Chain Visual Analytics

Graphs provide a more efficient representation of large analytical outputs.

Important visualisations include:

Visual analysis	Primary measure	Management purpose
Demand profile	Observed demand by medicine or period	Identify consumption patterns and concentration.
Inventory coverage	Stock relative to observed demand	Identify potentially excessive or constrained coverage.
ABC concentration	Revenue or consumption value by class	Show where economic attention is concentrated.
EOQ comparison	Current stock versus calculated EOQ	Compare current position with the model's economic reference quantity.
Expiry exposure	Quantity or value by expiry horizon	Visualise time-related inventory vulnerability.
Supplier analysis	Purchase quantity and procurement cost	Compare procurement activity and supplier contribution.
Risk distribution	Simulated or estimated outcomes	Show uncertainty rather than relying on a single point estimate.
Model performance	Observed versus predicted outcomes	Evaluate how well analytical models correspond to subsequent evidence.

Graphs should complement, rather than replace, the underlying numerical evidence.

1.7 Analytical Model Portfolio

The supply chain analytical portfolio includes:

Model	Question	Analytical output
Demand Forecasting	What demand may occur during the planning horizon?	Forecast and associated uncertainty.
ABC Analysis	Which medicines deserve greater management attention?	Priority classification and concentration analysis.
Stock Coverage	How long could current stock support observed demand?	Coverage estimate and exception signals.
Expiry Risk	Which inventory may become vulnerable before consumption?	Expiry exposure and priority signals.
Reorder Point	When should replenishment be considered?	Replenishment threshold based on demand and lead-time assumptions.
Safety Stock	How much uncertainty protection may be required?	Buffer estimate based on demand and supply variability.
EOQ	What quantity provides an economic reference under stated assumptions?	Calculated economic order quantity.
Monte Carlo	What outcomes are plausible under uncertainty?	Probability distributions and risk scenarios.

These models are complementary. No single model represents the entire supply chain.

1.8 Model Assessment and Validation

A mature supply chain intelligence system must evaluate its models rather than only execute them.

The conceptual assessment framework is:

Model Calculation → Validation → Performance Assessment → Calibration → Operational Use.

Model assessment may include:

- predictive accuracy;

- sensitivity and specificity where classification is appropriate;
- precision and recall;
- F1 score;
- calibration;
- threshold analysis;
- confusion matrices;
- residual analysis;
- forecast error measures;
- sensitivity analysis;
- scenario analysis;
- uncertainty assessment.

1.9 ROC Curve and Classification Assessment

ROC analysis is appropriate for supply chain models that classify a future binary event, such as whether an item will require a replenishment review, experience a stockout, or enter a defined risk category.

The receiver operating characteristic framework evaluates the relationship between:

True Positive Rate and False Positive Rate.

The area under the ROC curve (AUROC) provides a threshold-independent measure of discrimination.

However, ROC analysis should **not** be applied indiscriminately to continuous optimisation models.

For example:

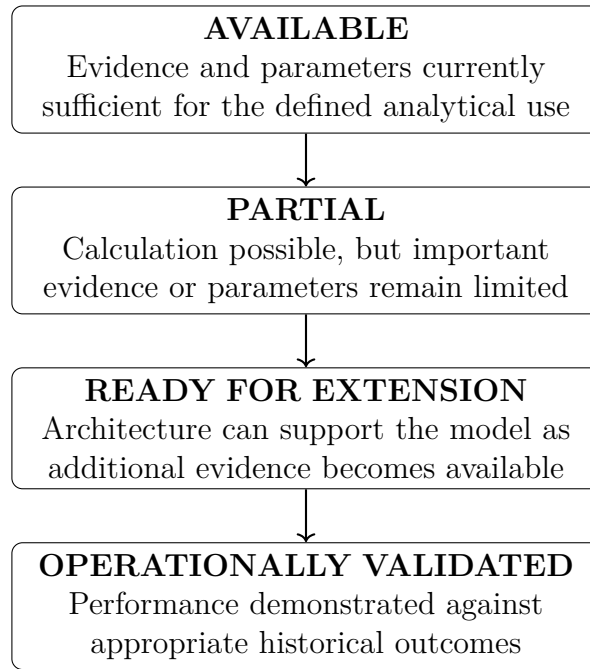
Model	Appropriate assessment approach
Demand Forecasting	Forecast error, residual analysis, calibration, and prediction intervals.
EOQ	Parameter sensitivity, cost assumptions, and scenario analysis.
ROP / Safety Stock	Service-level performance, stockout behaviour, and sensitivity to demand and lead-time assumptions.
Expiry Risk Classification	Confusion matrix, precision, recall, and ROC/AUROC where sufficient labelled outcomes exist.
Stockout Classification	ROC/AUROC, precision, recall, sensitivity, and specificity where historical outcomes are available.
Monte Carlo	Distribution validation, scenario sensitivity, and comparison of simulated versus observed outcomes.

Thus, **model assessment is model-specific**. The architecture should select the assessment method according to the type of prediction or decision being evaluated.

1.10 Model Readiness

Before a model becomes operationally influential, V13 should assess whether the necessary evidence and parameters are available.

A conceptual readiness classification is:



This prevents a mathematically valid calculation from automatically becoming an operational recommendation.

1.11 Management Interpretation

Supply chain intelligence should ultimately answer management questions rather than merely display model outputs.

A management interpretation layer may synthesise:

Demand + Stock + Expiry + Supplier + Financial Exposure + Model Evidence

to identify areas requiring attention.

For example:

High Expected Demand+Low Coverage+Long Lead Time $\Rightarrow$ Replenishment Attention.

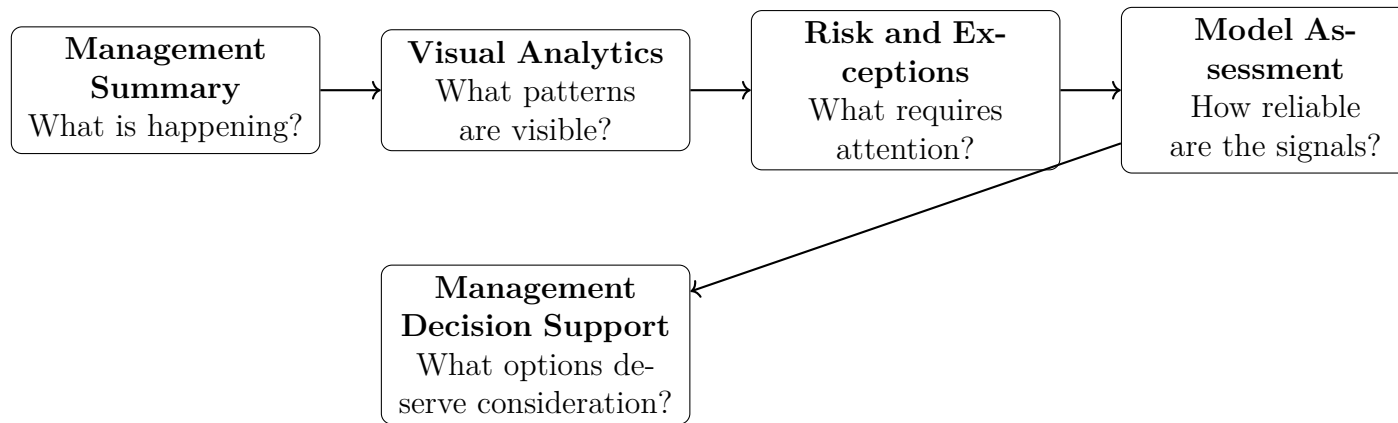
Conversely:

Low Expected Demand+High Stock+Short Remaining Shelf Life $\Rightarrow$ Expiry Exposure.

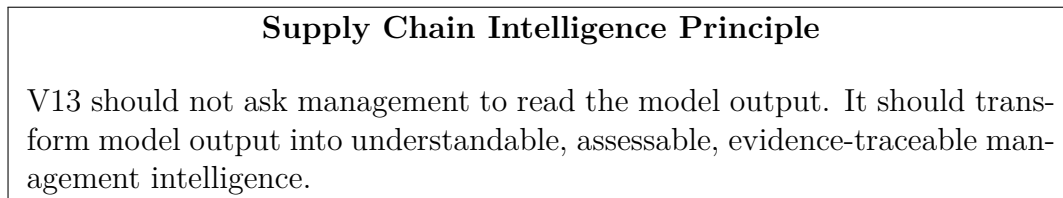
These expressions represent **management signals**, not automatic management decisions.

1.12 Supply Chain Management Cockpit

The final management-facing supply chain view should therefore move from detailed computational tables toward a layered cockpit:



The detailed Python model tables remain available as supporting analytical evidence, while the management interface presents their meaning in a compact, visual, and assessable form.



1.13 Future Development

As the historical evidence base grows, the supply chain layer can progressively move from exploratory analysis toward validated predictive and prescriptive decision support.

The intended progression is:

Observed Transactions → Reliable History → Validated Models → Performance Assessment → Risk-A

This provides a scientifically defensible path from the current operational system toward a mature pharmacy supply chain intelligence platform.